



Robotics

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Designing machines that sense, think, and act

Robotics combines mechanical engineering, electronics, and computer science to design machines that can sense their environment, make decisions, and perform physical tasks autonomously or semi-autonomously.

● DEFINITION

The interdisciplinary field involving the design, construction, and operation of robots that can perform tasks autonomously or with human guidance.

● WHY IT MATTERS

Robots increasingly handle dangerous, repetitive, or precision tasks across manufacturing, healthcare, and exploration, extending human capability significantly.

● WORKING PRINCIPLE



Robots use sensors to perceive their environment, processors to interpret data and make decisions, and actuators (motors) to perform physical actions, often guided by control algorithms that enable increasingly autonomous behavior.

● QUICK FACTS

- ▶ Surgical robots can perform procedures with sub-millimetre precision
- ▶ Robots have been sent to explore environments too dangerous for humans, from volcano interiors to Mars

● KEY TERMS

- Actuator
- Sensor
- Autonomous navigation
- Kinematics

● CORE CONCEPTS

- ▶ Sensors and actuators
- ▶ Robot kinematics
- ▶ Autonomous navigation
- ▶ Human-robot interaction

● APPLICATIONS

- ▶ Industrial manufacturing automation
- ▶ Surgical robotics
- ▶ Autonomous exploration (space, underwater)
- ▶ Warehouse and logistics automation

● ADVANTAGES

- ▶ Performs dangerous or repetitive tasks safely
- ▶ Improves precision beyond human capability in some tasks
- ▶ Increases manufacturing efficiency and consistency

CAREER OPPORTUNITIES

Robotics engineer, Automation engineer, Robotics software developer, Mechatronics specialist

RESEARCH AREAS

Autonomous navigation, Human-robot collaboration, Soft robotics

● REAL-LIFE EXAMPLES

- ▶ Robotic arms on car assembly lines
- ▶ Surgical robots assisting in operations
- ▶ Robotic vacuum cleaners

● LIMITATIONS

- ▶ High development and deployment costs
- ▶ Complex environments remain challenging for full autonomy
- ▶ Raises questions about workforce displacement

INDUSTRY APPLICATIONS

Manufacturing, Healthcare, Logistics and warehousing, Space and defense

FUTURE SCOPE

Advances in AI perception and control are rapidly expanding robots' ability to operate autonomously in unstructured, real-world environments.

● CURRENT TRENDS

Rapid growth of AI-powered humanoid and collaborative robots designed to work safely alongside humans.

● SUMMARY

Robotics designs machines that sense, decide, and act in the physical world, transforming manufacturing, healthcare, and exploration.

